

Bootstrap Inference in the Presence of Bias

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Toulouse School of Economics — Econometrics Reading Group

1. **Introduction** — the bootstrap's third role
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3. **General theory** — Theorem 3.1 and pre pivoting
4. **Examples revisited** — implementing the fix
5. **Wrap-up** — summary of the contribution

Introduction

Introduction

What we have seen so far:

- Bootstrap as a tool to compute the variance / distribution of a statistic when analytical formulas are unavailable.
- Bootstrap for asymptotic refinements: faster convergence than the first-order normal approximation (Beran 1987, Hall 1992).

Today — a different question:

- What if the estimator carries an asymptotic bias that the bootstrap simply **cannot** replicate?
- Then standard bootstrap inference is **invalid**, even asymptotically.
- Cavaliere et al. (2024): valid inference can still be restored without estimating the bias, by transforming the bootstrap p -value (*prepivoting*, Beran 1987, 1988).

Setup — the original statistic (1.1)

Scalar parameter θ , estimator $\hat{\theta}_n$ with rate $g(n) \rightarrow \infty$:

$$T_n := g(n)(\hat{\theta}_n - \theta) \xrightarrow{d} B + \xi_1, \quad (1.1)$$

with ξ_1 centred at zero, continuous cdf G .

- B : asymptotic bias; in the cases of interest **not consistently estimable**.
- Typical case: $g(n) = n^{1/2}$, $\xi_1 \sim N(0, \sigma^2)$.
- **Why classical inference fails:** quantiles of ξ_1 ignore B .

Why bootstrap, not analytical bias correction?

A consistent estimator of B often does not exist (model averaging, ridge), or exists but performs poorly in samples (nonparametric $\hat{\beta}''(x)$). Hall (1992), Horowitz (2001), Cattaneo et al. (2018, 2022): the standard bootstrap also *cannot* fix the bias when B cannot be estimated.

Setup — the bootstrap statistic (1.2)

Bootstrap statistic — centred at $\hat{\theta}_n$ (the bootstrap world's “truth”):

$$T_n^* := g(n)(\hat{\theta}_n^* - \hat{\theta}_n), \quad T_n^* - \hat{B}_n \xrightarrow{d^*}_p \xi_1. \quad (1.2)$$

- $\hat{B}_n$: implicit bootstrap bias — what the bootstrap DGP *produces*, not something the researcher chooses.
- Noise ξ_1 is replicated correctly under P^* .
- If $\hat{B}_n - B = o_p(1)$, bootstrap is valid in the usual sense (*).

Notation (used throughout)

$Y_n^* \xrightarrow{p}_p 0$ means $P^*(|Y_n^*| > \varepsilon) \xrightarrow{p} 0$ for every $\varepsilon > 0$.

$Y_n^* \xrightarrow{d^*}_p \xi$ means $P^*(Y_n^* \leq u) \xrightarrow{p} P(\xi \leq u)$ at every continuity point.

The failure mode (1.3)

$$\hat{B}_n - B \xrightarrow{d} \xi_2 \neq 0, \quad \text{jointly with (1.1).} \quad (1.3)$$

- The bootstrap bias is a random variable in the limit — it tracks B with irreducible noise ξ_2 .
- Hence the bootstrap law of T_n^* , conditional on D_n , is random in the limit: it cannot mimic the law of T_n .
- $\hat{p}_n := P^*(T_n^* \leq T_n)$ is **not asymptotically uniform**.
- Consequence: tests/CIs do not deliver nominal level (a nominal 5% test may reject at $\sim 15\%$).

Goal of the paper

Restore valid inference based on T_n when B is not consistently estimable — by working with the bootstrap p -value $\hat{p}_n$ directly, not by trying to estimate B .

A way out — pre pivoting

Idea (Beran 1987, 1988): let

$$H(u) := \lim_n \mathbb{P}(\hat{p}_n \leq u) \quad (\text{asymptotic cdf of the bootstrap } p\text{-value}).$$

Even though $H \neq U[0, 1]$, it can be:

- **continuous**, and
- **free of the unknown bias B** .

If so, the probability integral transform gives

$$H(\hat{p}_n) \xrightarrow{d} U_{[0,1]}.$$

Replace $\hat{p}_n$ by the modified p -value $\tilde{p}_n := \hat{H}_n(\hat{p}_n)$, where $\hat{H}_n \xrightarrow{P} H$ uniformly.

- Two implementations: **plug-in** (when $H = H_\gamma$ has finite-dim γ) and **double bootstrap** (estimate H nonparametrically).
- Beran proposed this for *refinements*; here the same device rescues a broken bootstrap.

Motivating Examples

Two motivating examples

Two concrete settings where (1.1)–(1.3) hold.

For each one we ask:

1. What is the model and the parameter of interest?
2. Does **(1.1)** hold? — $T_n \xrightarrow{d} B + \xi_1$ with $B \neq 0$.
3. Does **(1.2)** hold? — the bootstrap reproduces the noise ξ_1 .
4. Does **(1.3)** hold? — $\hat{B}_n - B \xrightarrow{d} \xi_2 \neq 0$. This is the condition that makes the standard bootstrap fail.

The two examples

1. **Ridge regression** with shrinkage tuned to the local-to-zero scale.
2. **Nonparametric regression** with MSE-optimal bandwidth.

A third example — inference after model averaging (Hansen 2007) — is in the paper; it shares the same mechanism as ridge.

Example 1: Ridge regression — setup

Linear regression $y_t = \theta' x_t + \varepsilon_t$, test $H_0 : g' \theta = r$ ($g \in \mathbb{R}^p$ fixed, scalar restriction).

Ridge estimator with shrinkage c_n :

$$\tilde{\theta}_n = \tilde{S}_{xx}^{-1} S_{xy}, \quad \tilde{S}_{xx} := S_{xx} + n^{-1} c_n I_p.$$

Local-to-zero setting: $\theta = \delta n^{-1/2}$ and $c_n/n \rightarrow c_0 > 0$.

Interesting regime: regressors carry modest information about θ ($\theta \sim n^{-1/2}$); calibrating $c_n \sim n$ puts shrinkage bias and sampling noise at the same order — the regime where the bias matters.

Example 1: Why (1.1)–(1.3) hold

(1.1) holds.

$$T_n = n^{1/2}(g'\tilde{\theta}_n - r) \xrightarrow{d} \underbrace{B}_{\propto c_0 \delta} + \underbrace{\xi_1}_{\sim N(0, v^2)}$$

Bias survives in the limit and depends on the *inestimable* δ .

(1.2) holds. Pairs bootstrap, centred at OLS $\hat{\theta}_n$: $T_n^* - \hat{B}_n \xrightarrow{d^*}_p N(0, v^2)$.

(1.3) holds — the bootstrap bias misses by $O_p(1)$

$$\hat{B}_n - B_n = -\frac{c_n}{n} g' \tilde{S}_{xx}^{-1} n^{1/2}(\hat{\theta}_n - \theta) \xrightarrow{d} \xi_2 \neq 0.$$

$c_n/n \rightarrow c_0 > 0$ and $n^{1/2}(\hat{\theta}_n - \theta) = O_p(1)$ **not** $o_p(1) \Rightarrow$ product is $O_p(1)$, irreducible.

Centre at OLS, not ridge, to preserve $E^*[x_t^* \varepsilon_t^*] = 0$ (GMM-bootstrap recentring).

Example 2: Nonparametric regression — setup

Nonparametric regression with smooth $\beta(\cdot)$:

$$y_t = \beta(x_t) + \varepsilon_t, \quad x_t = t/n, \quad \varepsilon_t \sim \text{iid}(0, \sigma^2).$$

Nadaraya–Watson estimator at fixed $x \in (0, 1)$ with kernel K :

$$\hat{\beta}_h(x) = \frac{1}{nh} \sum_{t=1}^n K\left(\frac{x_t - x}{h}\right) y_t, \quad h = c n^{-1/5}.$$

$h = c n^{-1/5}$ is the MSE-optimal rate: it balances squared bias against variance and gives the fastest rate of convergence. But at this h the bias does not vanish faster than the noise, so it shows up in the asymptotic distribution.

Example 2: Why (1.1)–(1.3) hold

(1.1) holds. Taylor expansion at x gives

$$T_n = (nh)^{1/2}(\hat{\beta}_h(x) - \beta(x)) = \underbrace{B_n}_{\propto \beta''(x)} + \underbrace{\xi_{1,n}}_{\xrightarrow{d} N(0, v^2)}.$$

Bias depends on the inestimable-in-practice $\beta''(x)$.

(1.2) holds (parametric bootstrap). $y_t^* = \hat{\beta}_h(x_t) + \varepsilon_t^*$, $\varepsilon_t^* \sim N(0, \hat{\sigma}_n^2)$; centring at $\hat{\beta}_h(x)$ gives $T_n^* - \hat{B}_n \xrightarrow{d^*}_p N(0, v^2)$.

(1.3) holds — different mechanism, same conclusion

The bootstrap bias $\hat{B}_n$ replaces $\beta(\cdot)$ by $\hat{\beta}_h(\cdot)$ — it re-smooths an already-smoothed curve. The relevant curvature error is

$$\hat{\beta}_h''(x) - \beta''(x) = O_p((nh)^{-1/2}),$$

exactly the same order as the bias itself at the MSE-optimal bandwidth. Hence

$$\hat{B}_n - B_n \xrightarrow{d} \xi_2 \neq 0.$$

Examples — recap

	Ridge	Nonparametric
Rate $g(n)$	$n^{1/2}$	$(nh)^{1/2} \approx n^{2/5}$
Bias B from	shrinkage to zero	smoothing curvature
Inestimable	δ (local to zero)	$\beta''(x)$ (impractical)
Bootstrap	pairs (centred at OLS)	parametric
Mechanism	$\hat{\theta}_n$ only $n^{1/2}$ -consistent	MSE-optimal h

One-line takeaway

Bias and noise live at the **same rate** $\iff (1.3) \iff$ bootstrap fails.

- Generic pattern, not a quirk of one model.

Now: a **general theory** that handles both at once.

General Results

Recap of notation (paper, Section 3)

- D_n : the original sample. D_n^* : a bootstrap sample drawn given D_n .
- $T_n = g(n)(\hat{\theta}_n - \theta)$: the original statistic; $T_n^* = g(n)(\hat{\theta}_n^* - \hat{\theta}_n)$ its bootstrap version.
- B_n : deterministic asymptotic bias of T_n (limit B).
- $\hat{B}_n$: D_n -measurable implicit bootstrap bias.
- $P^*(\cdot) = P(\cdot \mid D_n)$: bootstrap probability measure.

Modes of convergence (used throughout)

- $Y_n \xrightarrow{P} \xi$: convergence in probability (under P).
- $Y_n \xrightarrow{d} \xi$: convergence in distribution (under P).
- $Y_n^* \xrightarrow{P} 0$: $P^*(|Y_n^*| > \varepsilon) \xrightarrow{P} 0$ for all $\varepsilon > 0$.
- $Y_n^* \xrightarrow{d^*} \xi$: $P^*(Y_n^* \leq u) \xrightarrow{P} P(\xi \leq u)$ at every continuity point of the cdf.

G : cdf of ξ_1 . F : cdf of $\xi_1 - \xi_2$. $H = F \circ G^{-1}$ will be the cdf of $\hat{\rho}_n$.

Assumption 1 — the original statistic

Assumption (1)

$T_n - B_n \xrightarrow{d} \xi_1$, where ξ_1 is centred at zero with cdf

$$G(u) := \mathbb{P}(\xi_1 \leq u)$$

that is continuous and strictly increasing on its support.

Why these conditions on G ?

- **Continuous:** needed for Polya's theorem (pointwise $\Rightarrow$ uniform convergence of cdfs).
- **Strictly increasing:** ensures G^{-1} is well-defined, which we need for the probability integral transform $G(\xi_1) \sim U[0, 1]$.
- Not restricted to Gaussian — covers stable distributions, etc.

Assumption 2 — the bootstrap side

Assumption (2)

There exists D_n -measurable $\hat{B}_n$ such that:

- (i) $T_n^* - \hat{B}_n \xrightarrow{d^*}_p \xi_1$
- (ii) $\begin{pmatrix} T_n - B_n \\ \hat{B}_n - B_n \end{pmatrix} \xrightarrow{d} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}$, with $F(u) := \mathbb{P}(\xi_1 - \xi_2 \leq u)$ continuous.

- $\hat{B}_n$ is not a deliberate estimator of B_n — it is whatever bias the bootstrap DGP *implicitly* generates.
- Part (i): the bootstrap captures the noise correctly.
- Part (ii): joint convergence; ξ_2 quantifies the bias estimation failure.
- F must be continuous (but not necessarily strictly increasing) so that $H = F \circ G^{-1}$ is continuous.

Benchmark: what if $\hat{B}_n$ were consistent?

Suppose $\hat{B}_n - B_n = o_p(1)$, i.e. $\xi_2 = 0$ a.s. Then $F = G$ and:

$$\begin{aligned}\hat{p}_n &= P^*(T_n^* \leq T_n) = P^*(T_n^* - \hat{B}_n \leq T_n - \hat{B}_n) \\ &= G(T_n - \hat{B}_n) + o_p(1) \quad (\text{Assumption 2(i) + Polya}) \\ &\xrightarrow{d} G(\xi_1) \quad (\text{since } T_n - \hat{B}_n = T_n - B_n + o_p(1) \xrightarrow{d} \xi_1) \\ &\sim U[0, 1] \quad (\text{probability integral transform, using } F = G)\end{aligned}$$

Theorem 3.1 — the limit of $\hat{p}_n$

Theorem 3.1 (Cavaliere et al. 2024)

Under Assumptions 1 and 2,

$$\hat{p}_n \xrightarrow{d} G(F^{-1}(U_{[0,1]})).$$

Implications:

- $\hat{p}_n$ is **not uniform** unless $G = F$ (i.e. $\xi_2 = 0$).
- Cdf of the limit $V := G(F^{-1}(U))$:

$$\mathbb{P}(V \leq u) = \mathbb{P}(U \leq F(G^{-1}(u))) = F(G^{-1}(u)) =: H(u).$$

- H continuous and free of B (F, G are laws of centred objects).

Theorem 3.1 — proof

Step 1. Rewrite, subtracting $\hat{B}_n$ (constant under P^*):

$$\hat{p}_n = P^*(T_n^* - \hat{B}_n \leq T_n - \hat{B}_n)$$

By Assumption 2(i), $P^*(T_n^* - \hat{B}_n \leq u) \rightarrow G(u)$ uniformly (Polya). Hence:

$$\hat{p}_n = G(T_n - \hat{B}_n) + o_p(1)$$

Step 2. Decompose $T_n - \hat{B}_n = (T_n - B_n) - (\hat{B}_n - B_n)$. By Assumption 2(ii) and the continuous mapping theorem:

$$T_n - \hat{B}_n \xrightarrow{d} \xi_1 - \xi_2$$

Step 3. By continuity of G and continuous mapping: $G(T_n - \hat{B}_n) \xrightarrow{d} G(\xi_1 - \xi_2)$.

Step 4. Since $\xi_1 - \xi_2$ has continuous cdf F , the probability integral transform gives $\xi_1 - \xi_2 \stackrel{d}{=} F^{-1}(U)$. Substitute:

$$\hat{p}_n \xrightarrow{d} G(F^{-1}(U_{[0,1]})) \quad \square$$

Three remarks

Remark 3.1 — the most important

If a bootstrap scheme gives $\hat{B}_n \xrightarrow{P} 0$, then $\xi_2 = -B$ (constant) and H depends on B — prepivoting cannot work. Lesson: prefer schemes that try to capture the bias (even noisily).

- **Remark 3.2.** Confidence intervals are similarly invalid:
 $P(\theta_0 \in CI_n^{1-\alpha}) = P(\hat{p}_n \leq 1 - \alpha) \neq 1 - \alpha$.
- **Remark 3.3.** When ξ_2 is nondegenerate, the bootstrap conditional distribution $\hat{L}_n$ is random in the limit: $\hat{L}_n(u) \rightarrow G(u - B - \xi_2)$.

Corollary 3.1 — pre pivoting restores validity

Corollary 3.1

Under the conditions of Theorem 3.1, $H(\hat{p}_n) \xrightarrow{d} U_{[0,1]}$.

Proof. Let $V := G(F^{-1}(U_{[0,1]}))$ be the limit from Theorem 3.1. Apply H :

$$H(V) = F(G^{-1}(G(F^{-1}(U_{[0,1]})))) = F(F^{-1}(U_{[0,1]})) = U_{[0,1]}.$$

By continuous mapping, $H(\hat{p}_n) \xrightarrow{d} H(V) = U_{[0,1]}$. $\square$

Reading: applying H “unwarps” the distorted distribution of $\hat{p}_n$ back to uniform — exactly what the standard bootstrap was supposed to deliver.

But H depends on the unknown $F, G \implies$ we need an estimator $\hat{H}_n$.

Approach A: Plug-in

When $H = H_\gamma$ for some finite-dimensional γ , estimate γ and plug in.

Corollary 3.2

If $H_\gamma(u)$ is continuous in (γ, u) and $\hat{\gamma}_n \xrightarrow{P} \gamma$, then

$$\tilde{p}_n := H_{\hat{\gamma}_n}(\hat{p}_n) \xrightarrow{d} U_{[0,1]}.$$

Proof. Two pieces.

1. By joint continuity of H_γ in γ , $\sup_u |H_{\hat{\gamma}_n}(u) - H_\gamma(u)| \xrightarrow{P} 0$. Hence $\tilde{p}_n - H(\hat{p}_n) \xrightarrow{P} 0$.
2. By Corollary 3.1, $H(\hat{p}_n) \xrightarrow{d} U_{[0,1]}$.

Slutsky $\Rightarrow \tilde{p}_n \xrightarrow{d} U_{[0,1]}$. $\square$

Cheap (one bootstrap + one formula), but requires deriving H_γ in closed form. In the Gaussian case — which covers our examples — it takes a simple shape: see next slide.

Gaussian plug-in — the correction m

$\xi_1 \sim N(0, v^2)$, (ξ_1, ξ_2) jointly normal, $\xi_1 - \xi_2 \sim N(0, v_d^2)$.

$$\begin{aligned} G(u) &= \Phi(u/v), & F(u) &= \Phi(u/v_d), & G^{-1}(u) &= v \Phi^{-1}(u). \\ H(u) &= F(G^{-1}(u)) = \Phi\left(\frac{v \Phi^{-1}(u)}{v_d}\right) = \Phi(m^{-1} \Phi^{-1}(u)), & m &:= v_d/v. \end{aligned}$$

That is:

- Under (1.3): $T_n - \hat{B}_n \xrightarrow{d} N(0, v_d^2)$ — the actual spread is $v_d^2 \neq v^2$.
- $\tilde{p}_n = \Phi(m^{-1} \Phi^{-1}(\hat{p}_n))$ rescales the p -value as if the spread had always been v_d^2 . B drops out completely.

In our examples

Ridge: m^2 depends on Σ_{xx}, c_0 — not on the unknown δ .

Nonparametric: m^2 depends only on the kernel K (σ^2 cancels) — no estimation at all.

Approach B: Double bootstrap

Construction:

1. From each bootstrap sample D_n^* , draw a second-level sample D_n^{**} .
2. Compute T_n^{**} and the second-level p -value $\hat{p}_n^* := P^{**}(T_n^{**} \leq T_n^*)$.
3. Estimate H by $\hat{H}_n(u) := P^*(\hat{p}_n^* \leq u)$.
4. Modified p -value: $\tilde{p}_n := \hat{H}_n(\hat{p}_n) = P^*(\hat{p}_n^* \leq \hat{p}_n)$.

Assumption 3 ensures the double bootstrap replicates the distortion:

- (i) $T_n^{**} - \hat{B}_n^* \xrightarrow{d^{**}}_{P^*} \xi_1$ (noise replicated at level 2)
- (ii) $T_n^* - \hat{B}_n^* \xrightarrow{d^*}_P \xi_1 - \xi_2$ (bias gap replicated)

Theorem 3.2

Under Assumptions 1, 2, 3: $\tilde{p}_n = \hat{H}_n(\hat{p}_n) \xrightarrow{d} U_{[0,1]}$.

Corollary 3.3 — the practical shortcut

Corollary 3.3

Under Assumptions 1, 2, 3:

$$\tilde{p}_n = P^*(T_n^* - \hat{B}_n^* \leq T_n - \hat{B}_n) + o_p(1).$$

the double bootstrap modified p -value is asymptotically equivalent to a single-level bootstrap applied to the bias-corrected statistic $T_n - \hat{B}_n$.

Practical recipe:

1. Compute $\hat{B}_n$ from the original data (available in closed form for ridge, model averaging).
2. Compute $\hat{B}_n^*$ from each bootstrap sample.
3. Bootstrap p -value: $P^*(T_n^* - \hat{B}_n^* \leq T_n - \hat{B}_n)$.

No double bootstrap needed when $\hat{B}_n, \hat{B}_n^*$ are analytically available.

Examples Revisited

Ridge regression — implementing the fix

Lemma 4.4 verifies Assumptions 1–2 with $(\xi_1, \xi_2) \sim N(0, V)$, V depending on $\sigma^2, \Sigma_{xx}, c_0$ (all estimable). The distortion simplifies to:

$$m^2 = \frac{g' \Sigma_{xx}^{-1} g}{g' \tilde{\Sigma}_{xx}^{-1} \Sigma_{xx} \tilde{\Sigma}_{xx}^{-1} g}$$

Plug-in:

1. Compute ridge $\tilde{\theta}_n$ and $T_n = n^{1/2}(g' \tilde{\theta}_n - r)$.
2. Run pairs bootstrap: resample (y_t^*, x_t^*) , centre at OLS: $T_n^{*(b)} = n^{1/2} g'(\tilde{\theta}_n^{*(b)} - \hat{\theta}_n)$.
Compute $\hat{\rho}_n$.
3. $\hat{m}_n^2 = (g' S_{xx}^{-1} g) / (g' \tilde{S}_{xx}^{-1} S_{xx} \tilde{S}_{xx}^{-1} g)$.
4. Report $\tilde{\rho}_n = \Phi(\hat{m}_n^{-1} \Phi^{-1}(\hat{\rho}_n))$.

Lemma 4.5 verifies Assumption 3 (double bootstrap also valid).

Nonparametric regression — implementing the fix

Lemma 4.6 verifies Assumptions 1–2. Remarkable: σ^2 cancels in $m^2 = v_d^2/v^2$, so m depends only on the kernel K — known to the researcher, no estimation needed.

Plug in:

1. Compute $\hat{\beta}_h(x)$ with MSE-optimal $h = cn^{-1/5}$ and $T_n = (nh)^{1/2}(\hat{\beta}_h(x) - \beta(x))$.
2. Run parametric bootstrap: $y_t^* = \hat{\beta}_h(x_t) + \varepsilon_t^*$, $\varepsilon_t^* \sim N(0, \hat{\sigma}_n^2)$. Compute $\hat{p}_n$.
3. Look up m from K (a pre-computed constant).
4. Report $\tilde{p}_n = \Phi(m^{-1} \Phi^{-1}(\hat{p}_n))$.

Comparison: *undersmoothing* removes bias but kills local power; *robust bias correction* (CCF) needs a second bandwidth. This paper: keep MSE-optimal h , one-line correction, no $\beta''(x)$.

Lemma 4.7 verifies Assumption 3 for the double bootstrap (with a small observable correction to T_n^{**}).

Wrap-up

Summary

- **Problem.** When the estimator has an asymptotic bias B that the bootstrap cannot replicate (1.3), the bootstrap distribution is random in the limit and $\hat{p}_n$ is not uniform.
- **Diagnosis.** Bias and noise live at the same rate; an inestimable quantity (δ in model averaging / ridge, $\beta''(x)$ in nonparametrics) sits exactly at the boundary.
- **Fix — prepivoting.** The asymptotic cdf $H = F \circ G^{-1}$ of $\hat{p}_n$ is continuous and free of B ; transform $\tilde{p}_n = \hat{H}_n(\hat{p}_n) \xrightarrow{d} U_{[0,1]}$.
- **Two routes.** Plug-in (Gaussian: ratio of variances $m^2 = v_d^2/v^2$, no B) or double bootstrap (no closed form for H needed; Corollary 3.3 gives a single-bootstrap shortcut).

Take-away for the reading group

A new, third role for the bootstrap: *not* variance, *not* refinement — but [rescuing inference](#) when the first-order bootstrap is invalid, without ever estimating the bias.

Appendix: Gaussian special case

When $\xi_1 \sim N(0, v^2)$ and $(\xi_1, \xi_2) \sim N(0, V)$:

$$G(u) = \Phi(u/v), \quad F(u) = \Phi(u/v_d), \quad v_d^2 = v_{11} + v_{22} - 2v_{12}$$

$$H(u) = \Phi(m^{-1}\Phi^{-1}(u)), \quad m = v_d/v$$

Modified p -value: $\tilde{p}_n = \Phi(\hat{m}_n^{-1}\Phi^{-1}(\hat{p}_n))$.

Covers Examples 1–3 and Example 5 (panel). Example 4 (heavy tails) uses $G(u) = \Psi_\alpha(\omega^{-1}u)$ with Ψ_α the stable cdf.

Appendix: Section 3.5 — random centering (pairs bootstrap)

Some bootstrap schemes (e.g. pairs in model averaging) give $T_n^* - B_n^* \xrightarrow{d^*}_p \xi_1$ where B_n^* depends on bootstrap data.

Assumption 4 replaces 2(i): $T_n^* - \hat{B}_n \xrightarrow{d^*}_p \zeta_1$ with cdf $J \neq G$.

Theorem 3.4: $\hat{p}_n \xrightarrow{d} J(F^{-1}(U))$ and $H(u) = F(J^{-1}(u))$.

Plug-in and double bootstrap go through with J replacing G .

Caution: the plug-in formula for m is bootstrap-scheme-specific. If you derived m for FRB and applied it to pairs, you'd get the wrong correction. The double bootstrap handles this automatically.

Appendix: Why prepivoting, not studentisation?

Q. Why not just bootstrap the studentised statistic $T_n/\hat{s}e_n$?

A. Studentisation fixes *scale*, not *location*.

- Distortion here is $\hat{B}_n - B_n \xrightarrow{d} \xi_2$ — a **random shift**.
- Dividing by $\hat{s}e_n$ does not subtract a random location.
- Studentisation $\Rightarrow$ refinements when bias is $o_p(1)$; useless when bias is $O_p(1)$ relative to noise.

Appendix: Power (Theorem 3.3)

Test $H_0 : \theta = \bar{\theta}$ vs. $H_1 : \theta < \bar{\theta}$.

Under local alternatives $\theta_0 = \bar{\theta} + a/g(n)$ with $a < 0$:

$$P(H(\hat{p}_n(\bar{\theta})) \leq \alpha) \rightarrow F(F^{-1}(\alpha) - a)$$

- Fixed alternatives ($a \rightarrow -\infty$): test is consistent.
- Local alternatives: power depends on F only, not G .
- Same power as an oracle test that knows B .

Appendix: Supplement examples

Example 4 — Heavy tails: $y_t = \theta + \varepsilon_t$ with $\varepsilon_t \sim S(\alpha)$, $\alpha \in (1, 2)$ (infinite variance).

- Non-Gaussian ξ_1 , unknown convergence rate $n^{1-1/\alpha}$.
- Plug-in works: $H(u) = \Psi_\alpha(\omega \Psi_\alpha^{-1}(u))$.
- m -out-of- n bootstrap fails (zeros out bias $\Rightarrow$ Remark 3.1 applies).

Example 5 — Dynamic panel: Fixed effects MLE with incidental parameter bias.

- Cross-sectional pairs bootstrap gives $\hat{B}_n = 0 \Rightarrow$ distortion depends on B (Remark 3.1).
- But B can be estimated consistently here (Hahn–Kuersteiner 2011).
- Plug-in works; double bootstrap fails (Remark B.4: both levels zero out the bias).